

L12 — Autonomous Agents

Agent properties, dialogue trees, and knowledge graphs

Study summary generated from lecture slides. ETH AI for Digital Characters.

Agent definition and properties

- Agent: perceives environment via sensors, acts via actuators
- Reactivity: timely response to environmental changes
- Autonomy: operates without direct human control
- Proactiveness: plans for future opportunities (not just reacting)
- Socially intelligent agents: real-time perception, emotion, action in social settings

Decision-making approaches

- Finite State Machines: states + transition conditions
- Dialogue trees: hierarchical choices; low flexibility, hard to scale
- LLM-based: task decomposition, tool use, continuous interaction

Knowledge graphs

- RDF triples: (subject, predicate, object)
- Schema design with typed entities and relations
- SPARQL: W3C standard query language for structured retrieval
- Query types: one-hop, path, conjunctive

Knowledge graph embeddings

- TransE: $h + r \sim t$; score = $||h + r - t||$
- TransE limitation: symmetric relations ($r \sim 0$)
- TransR: entities and relations in different spaces
- Link prediction: infer missing edges in large/incomplete graphs
- Query2box: box embeddings for complex conjunctive queries

LLM + KG integration

- KG grounds LLM facts; reduces hallucination
- Use cases: smart home coordination, healthcare, recommendations, bias mitigation